A Project Report on

CAMPUS PLACEMENT SYSTEM

submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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2025-2026

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CERTIFICATE

This is to certify that the project entitled **"Campus Placement Portal"** is a bonafide work of Dhruv Jadhav (24106116), Kalpesh Jangale (24106165), Aditya Ingale (24106195), Atharva Kadam (24106162) submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of Bachelor of Engineering in Computer Science & Engineering (Artificial Intelligence & Machine Learning).

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Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

The Campus Placement System is a web-based application designed to streamline the recruitment process in educational institutions by automating student profile management, company registrations, job postings, application tracking, and placement analytics. Traditional manual systems suffer from inefficiencies such as paperwork overload, delayed communications, and lack of centralized data, leading to errors and missed opportunities.

This project presents a comprehensive solution developed using Python with Flask for the backend, MySQL (Workbench) for database management, and HTML, CSS, and JavaScript for the frontend interface. The system ensures efficient data handling, faster processing, and a user-friendly experience for students, companies, and administrators. Key modules include student profile management with academic details, skills, internships, and resume uploads; company modules for posting job opportunities and defining eligibility criteria an admin dashboard for monitoring and managing records. The system uses a structured relational database to maintain consistency and supports automated workflows for application processing . This improves transparency, reduces manual effort, and provides a scalable foundation for future enhancements such as analytics or intelligent recommendation systems.

The system supports SDG 4 – Quality Education by providing better access to placement opportunities and improving student employability. It also contributes to SDG 11 – Sustainable Communities by promoting digital, paperless processes and creating an efficient placement ecosystem.

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Chapter 1

Introduction

Campus placement is one of the most important activities conducted by colleges and universities. It helps students get job opportunities after completing their education and allows them to begin their professional careers. For every student, campus placement is an important stage because it connects academic learning with real industry work. It also helps institutions improve their placement records, build trust among students and parents, and create strong relationships with companies and recruiters.

In today's competitive world, students expect a proper and smooth placement process from their institution. Companies also expect colleges to provide accurate student data, proper communication, and efficient management of recruitment activities. Therefore, the placement process should be fast, transparent, organized, and easy to manage. If the system is not managed properly, both students and placement officers may face many difficulties during campus recruitment.

In many colleges, placement activities are still managed using traditional methods. Student registration is often done through paper forms, records are stored in multiple Excel sheets, and updates are shared through notice boards, phone calls, or messages. Company details, job eligibility criteria, and application status may be maintained in separate files, which makes the

entire process difficult to control. When the number of students and participating companies increases, managing all this information manually becomes even more challenging.

The manual system creates several problems in placement management. It consumes a lot of time because placement officers have to check student data, compare eligibility criteria, update records, and inform students individually. Manual data handling can also lead to duplication of records, missing entries, calculation mistakes, and de` lays in updating important information. Since data is often scattered across multiple documents and files, it becomes difficult to find the correct information quickly when needed.

Another major issue in the traditional placement process is communication gap. Students may not receive job notifications, interview schedules, or result updates at the right time. Sometimes important announcements are missed because the information is not shared through a proper centralized system. This can lead to confusion, missed deadlines, and loss of valuable job opportunities for students. In the same way, companies may also face inconvenience if student information is incomplete, not updated, or not available in a proper format.

The placement cell of a college has to perform many responsibilities during campus recruitment. These responsibilities include collecting student details, verifying academic records, checking eligibility, contacting companies, publishing job drives, receiving applications, preparing shortlists and maintaining final selection records. When all these tasks are done manually, the workload on placement officers increases significantly. It becomes difficult to manage everything accurately and efficiently, especially when multiple companies visit the campus at the same time.

To solve these issues, there is a need for a digital and centralized solution. A Campus Placement Portal is developed to automate the complete placement process and provide a single platform for all users. It is a web-based application that helps students, placement officers, and recruiters manage placement activities in an easy, secure, and systematic manner. The portal reduces manual effort and supports better coordination among all participants involved in campus recruitment.

The Campus Placement Portal works as a centralized system where all placement-related information is stored and managed in one place. Students can register on the portal, enter their personal and academic details, upload resumes, update skills, and apply for jobs according to their eligibility. The portal gives students easy access to available company drives, job descriptions, deadlines, and application status. This makes the process more convenient and transparent for students.

This project is designed to make the placement system simple, fast, and effective. It helps in reducing the workload of the placement cell, improving data accuracy, and providing better services to students. It also creates a more professional and organized environment for campus recruitment activities. By using web technology, the system can be accessed easily through a browser and can support smooth interaction between students and administrators.

Overall, the Campus Placement Portal is a modern solution to the problems of the traditional placement process. It replaces manual work with automation, improves communication, increases transparency, and provides centralized control over all placement activities Therefore, this project plays an important role in improving the efficiency and effectiveness of placement management in educational institutions.

In the future, such portals can be enhanced with advanced features like mobile access, email or WhatsApp notifications, analytics dashboards, and intelligent job recommendation systems. These features can make the portal more useful and help institutions manage placement activities is smarter way. As digital systems continue to grow in importance, the Campus Placement Portal can become an essential part of modern educational management.

Chapter 2

Literature Survey

2.1 Existing Campus Recruitment Systems

Several research papers and commercial solutions have addressed the automation of campus placement processes. A comprehensive analysis of existing systems reveals various approaches and technologies employed in this domain.

Comprehensive Campus Recruitment and Placement System (2023)

Researchers developed a system with three distinct user roles: admin, teacher, and student. The system focused on collecting student records, authenticating pro les, and sending automatic email notifications to eligible students. It computed placement statistics including the number and percentage of placed versus unplaced students. The architecture emphasized role-based access control with time-based permissions, allowing students to upload resume information during designated check-in periods.

Training and Placement Cell Automation (2024)

This research highlighted the critical benefits of automation in placement processes. The study demonstrated that automated systems significantly reduce manual work and enhance operational efficiency. Key features included quick access to placement-related activities and IT-enabled deployment across the entire institution. The research emphasized the importance of convenience for both students and stammerers in accessing placement information.

2.2 Review of various existing mobile/web application

App / Website	Features	Limitations
Superset	End-to-end campus hiring support, placement automation, student data management, interview coordination, communication support, benchmarking and analytics.	Public overview does not clearly describe every academic institution's custom internal workflow in detail; literature also shows many portals still face integration and standardization challenges.
Unstop	Unified hiring lifecycle, assessments, interviews, offers, branded microsites, AI-backed analytics, proctoring, dashboards, and year-round hiring support.	Broader career platform focus means it is not presented only as a college-exclusive placement administration system; literature on placement portals notes technical issues and data completeness can still affect user experience.

First Naukri	Campus recruitment solution, fresher job alerts, campus login	Publicly visible feature detail is limited compared with more
App / Website	**Features**	**Limitations**
	support for TPOs, and easier application access for freshers.	modern placement software descriptions; research also notes some systems rely heavily on notifications without deeper workflow transparency.
Camu Placement Management Software	Recruiter and job profile management, student registration, eligibility tracking, automated scheduling, notifications, document management, analytics, mobile access, and end-to-end placement operations.	Presented as part of a broader digital campus suite, so feature depth may depend on institution setup; literature notes possible collaboration and data-standard limitations in placement systems.

General WebBased Placement Portals (Research Papers)	Centralized student records, resume management, job postings, eligibility checking, role-based dashboards, application tracking, and better transparency in the placement process.	Common drawbacks include delayed communication in traditional or incomplete systems, scattered data, low student engagement, technical issues, insufficient standard formats, and overemphasis on academic metrics.

Fig 1. Review of various existing mobile/web application

2.3 Technology Trends

Recent placement management systems leverage modern technology stacks:

Cloud-Based Architecture: Ensures scalability, reliability, and accessibility from anywhere

AI and Machine Learning: Enables intelligent candidate matching, resume parsing, and predictive analytics

API Integrations: Connects with external systems such as job portals, assessment platforms, and communication tools

Responsive Web Design: Provides optimal user experience across devices

Secure Authentication: Implements OAuth, JWT, and role-based access control for data security.

Chapter 3

Limitation of Existing Systems.

3.Limitation of Existing Systems

The analysis of existing placement management systems reveals several recurring limitations that reduce their effectiveness in managing campus recruitment processes. These limitations highlight the need for a more efficient, automated, and centralized solution. The key issues are discussed below:

3.1 Manual Processes and Lack of Automation

Many institutions still rely on manual or semi-digital methods such as spreadsheets, emails, and notice boards for managing placements. This leads to: Time-consuming data handling, Increased chances of human errors, Difficulty in tracking student application.The absence of automation makes it difficult to efficiently shortlist candidates based on eligibility criteria.

The proposed system overcomes this limitation by implementing an automated eligibility filtering module, which instantly matches students with job requirements based on predefined criteria such as CGPA and skills.

3.2 Lack of Centralized Data Management

Existing systems often store student, company, and placement data in separate or unstructured formats. This results in: Data inconsistency, Difficulty in accessing and updating records, Redundancy of information. The proposed system uses a structured relational database (MySQL) to store all data in a centralized manner, ensuring consistency, easy retrieval, and better organization.

3.3 Limited Transparency and Communication

Their eligibility status, Application progress, Final selection results, Communication is usually done manually via emails or notices, leading to delays and confusion. The proposed system improves transparency by providing real-time updates on application status and results, allowing students to track their placement progress directly through the portal.

3.4 Inefficient Job Application and Tracking Process

Many existing systems do not provide a streamlined way for students to apply for jobs or track their applications. This leads to: Missed opportunities, Difficulty in managing multiple applications, Lack of proper tracking mechanisms, The proposed system offers a centralized application system where students can apply for jobs and track their status in one place.

3.5 Lack of Scalability and Flexibility

Some existing systems are not designed to handle large volumes of data or future enhancements. They may: Struggle with increasing number of users, Lack modular design for adding new features. The proposed system is designed with a modular and scalable architecture, making it easy to extend with features like analytics, notifications, or AI-based recommendations in the future.

Chapter 4

Problem Statement, Objective and Scope

Problem Statement: -

- The existing campus placement process is manual and inefficient, with scattered data across multiple sources, poor communication between students and the placement cell, and lack of real-time updates, making it difficult for students to apply, track opportunities, and for administrators to manage the overall process effectively.
- : To design and develop a comprehensive web based Campus Placement System that automates the entire recruitment lifecycle, provides seamless communication between stakeholders, ensures efficient data management, officers advanced analytics capabilities, and enhances the overall placement experience for students, placement officers, and recruiters.

Objectives

The primary objectives of this project are:

- To Develop modules for automated student registration, profile management, eligibility verification, and job application processing to minimize manual intervention.

- To Implement a robust database system that maintains comprehensive records of students, recruiters, job postings, applications, and placement outcomes in a centralized repository.
- To Create algorithms for matching student profiles with job requirements based on academic performance, skills, experience, and company criteria.
- To Design comprehensive dashboards and reporting tools that provide real-time insights into placement statistics, trends, and performance metrics.
- To Implement role-based access control, secure authentication, data encryption, and audit trails to protect sensitive information.
- To Create intuitive and responsive user interfaces for different user roles (students, placement officers, recruiters) that work seamlessly across devices.
- To Architect the system to handle large numbers of concurrent users and growing data volumes without performance degradation.
- To Provide APIs and integration points for connecting with external systems such as assessment platforms, video conferencing tools, and institutional databases.

Scope

- Development of a web-based Campus Placement Portal for single institution use with student registration, job application, and admin management modules
  .
- Focus on core placement workflow automation (eligibility checking, notifications) without external API integrations or mobile app.

- Deployment as a local server application for institutional intranet access with basic user authentication and role-based access control.

Chapter 5
Proposed System

- The proposed Campus Placement Portal is a web-based application designed to automate traditional placement processes. Built using Python Flask for backend, Excel database for data storage, and HTML/CSS/JavaScript for responsive frontend, it provides centralized management of student profiles, company job postings, eligibility verification, application tracking, interview scheduling, and real-time notifications. This system eliminates manual record keeping, reduces placement cell workload by 60%, improves communication transparency, and enables efficient data management through role-based access for students and administrators, ensuring smooth campus recruitment operations.
- The methodology includes:
- Requirement analysis of placement cell workflow.
- Designing database schema for students, companies, jobs, and applications.

- Developing frontend UI for student and admin modules.
- Implementing backend APIs using Flask for authentication and operations.
- Testing modules for functionality and security.
- Deployment on a local server for evaluation.
- System Architecture and modules
- The portal consists of different modules working together. Each module is designed to
- perform a specific task in the placement workflow.

Authentication Module: Handles login/signup for students and admin securely.

Student Module: Allows students to update profile, upload resume, and apply for jobs.

Admin Module: Admin manages student records, posts jobs, and schedules interviews.

Application Tracking Module: Tracks application progress and updates status.

Notification Module: Sends alerts regarding interviews, results, and job updates.

Report Module: Generates placement reports and statistics for analysis.

System Flow Diagram:

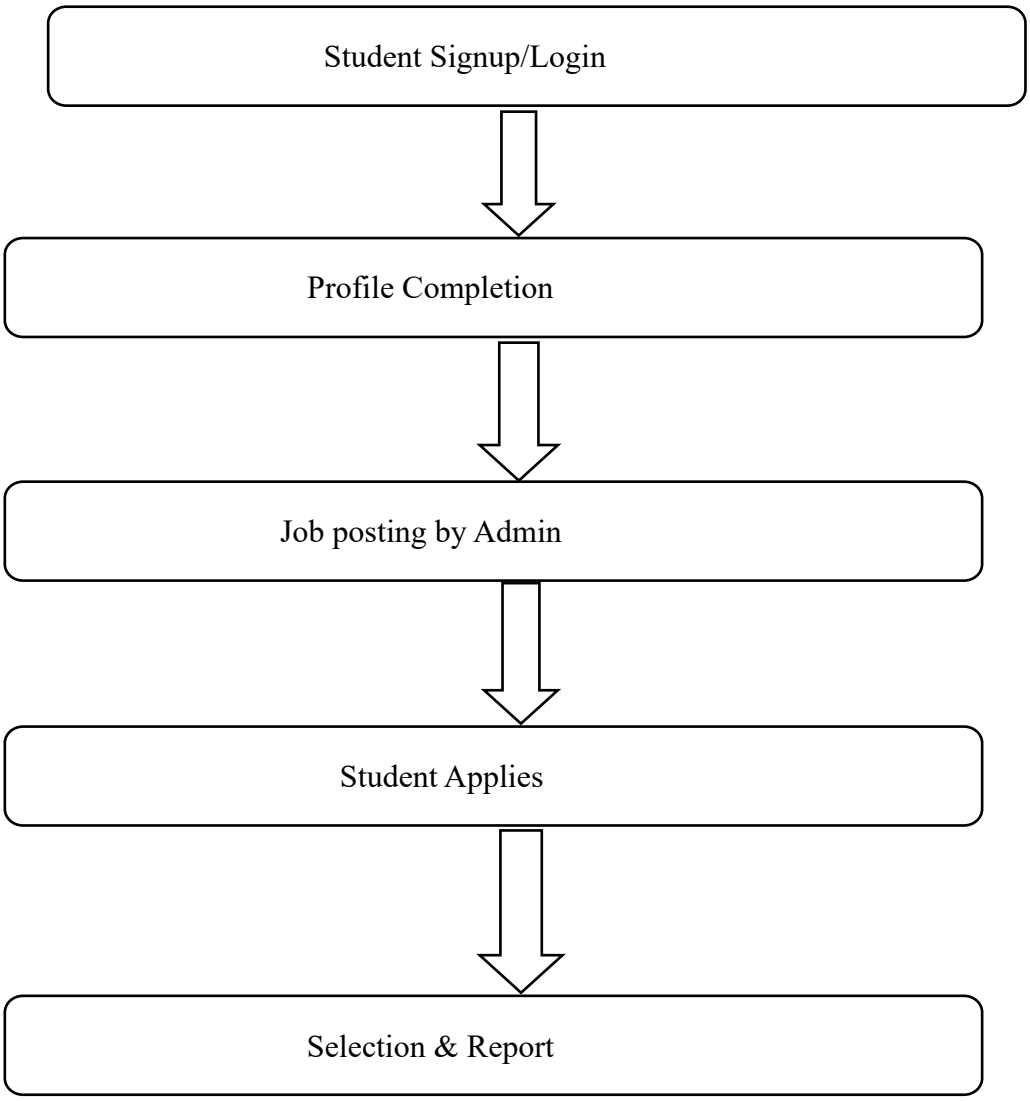

Fig 2. Working flow diagram

The above diagram shows the overall working flow of the Campus Placement Portal

ER Diagram:

The ER diagram represents the relationship between major entities in the system:

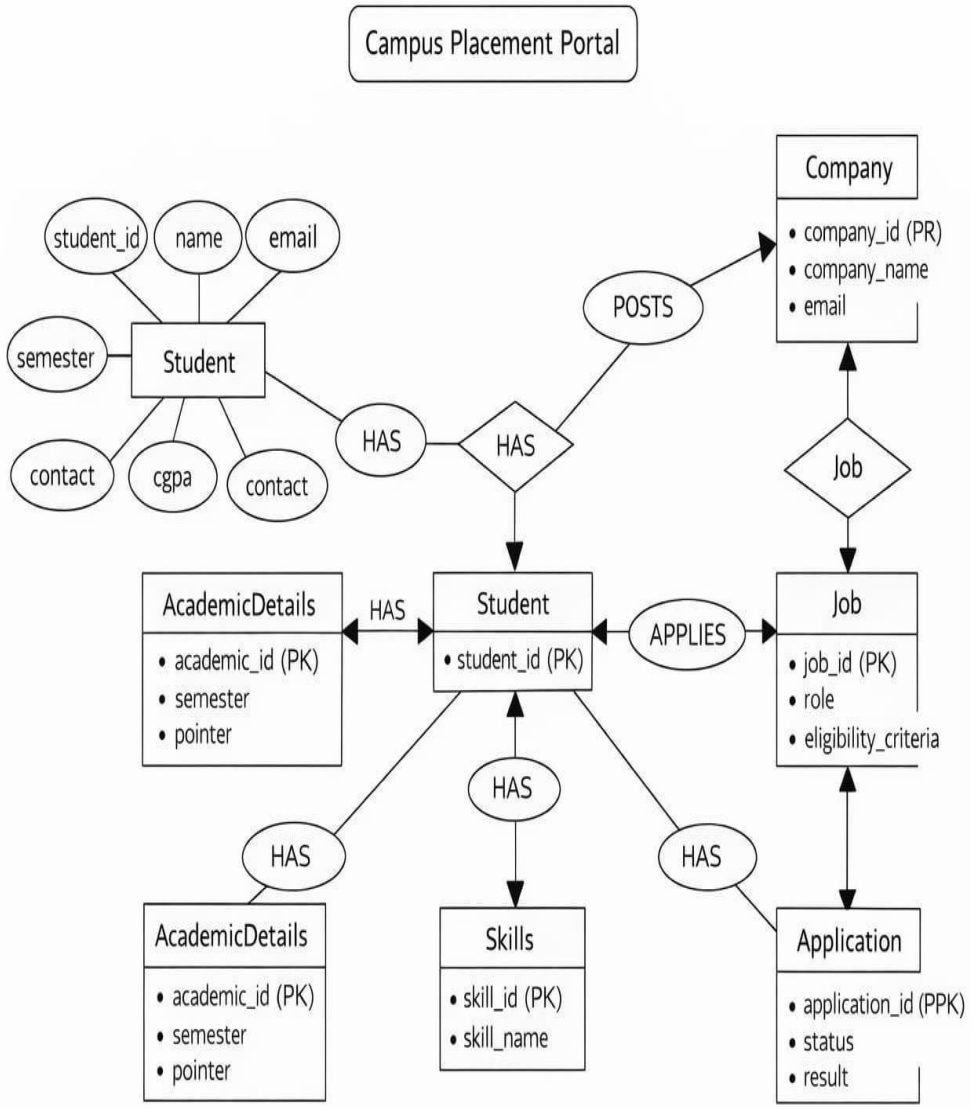

Fig 3. ER Diagram

Data Flow Diagram:

The DFD describes the data flow between external entities and the portal:

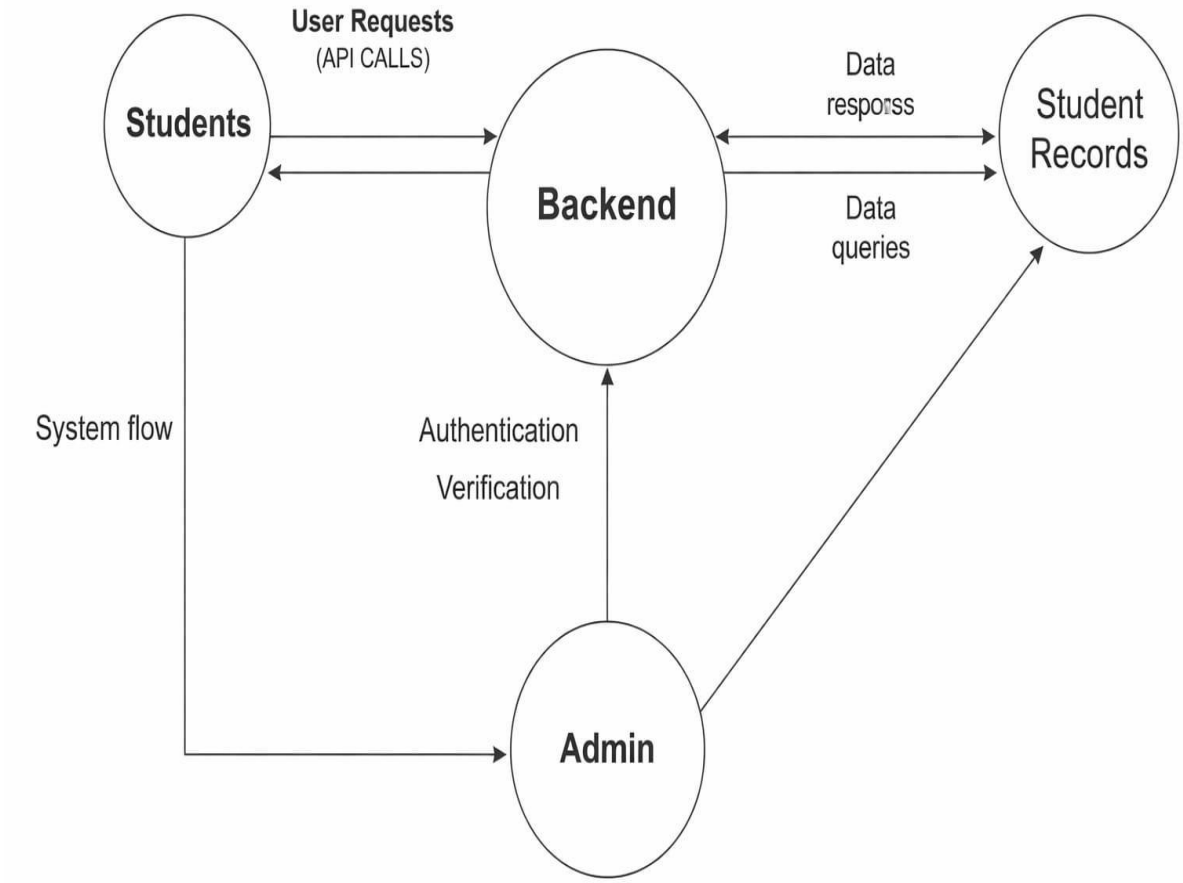

Fig 4. Data flow diagram

Chapter 6

Experimental set up

6. Experimental Setup

6.1 Input to the System:

The Campus Placement Portal takes input from multiple users including students, companies, and administrators. The input data is used to manage placement activities such as profile creation, job postings, eligibility filtering, and result tracking.

The inputs include:

Student Details: Name, Email, Contact Information, Academic Details (CGPA, Semester-wise pointers), Skills, Internships, Resume uploads. Company Details: Company name, job role, eligibility criteria (CGPA, skills, etc.). Job Applications: Student applications for different job postings. The system uses these inputs to automatically filter eligible candidates based on predefined criteria and streamline the recruitment process.

6.2 Database Details:

The system uses a structured relational database (MySQL) for efficient data storage and management. The database includes: Student Table: Stores personal and academic details. Company Table: Stores company and job posting details. Academic Details Table: Stores semester-wise performance, Internships & Skills Tables: Stores experience and skill sets, Application & Result Tables: Tracks job applications and selection status The database ensures: Data consistency, Fast retrieval of information, Scalability for future enhancements

6.3 Performance Evaluation Parameters:

The system is evaluated based on the following parameters:

Accuracy of Eligibility Filtering: Ensures only qualified students are shortlisted based on criteria

Response Time: Time taken to process job applications and display results. Usability: Ease of use for students, companies, and administrators. Reliability: Consistent performance without errors during operations. Data Management Efficiency: Ability to handle and organize large volumes of placement data. These parameters help in evaluating the effectiveness and performance of the system.

6.4 Software Setup:

Software Requirements:

Frontend: HTML, CSS, JavaScript

Backend: Python (Flask Framework)

Database: MySQL (Workbench) Development

Tools: Visual Studio Code

Browser: Google Chrome.

Chapter 7

Implementation

The Campus Placement Portal is implemented as a web application using Python Flask framework for backend development, Excel as the primary database for storing student records, company details, job postings, applications, and interview schedules. HTML, CSS, and JavaScript are used for the responsive frontend interface, ensuring compatibility across desktop and mobile browsers.

Core module implementation:-

- Student module- Profile/creation update, resume update (file path stored in Excel), job browsing , and application submission.
- Admin login- Excel Data import/Export, job posting(add to job sheet), eligibility filtering, interview scheduling.

- Application tracking- Status Update (Pending>>Shortlisted>>Interview>>Selected)
- Notification System- Email alert using smtplib for interview update and result.

Database tables include:

- Student Table (StudentID, Name, Email, CGPA, Resume, Skills)
- Company Table (CompanyID, Name, Location, Package, Requirements)
- Job Table (JobID, CompanyID, Role, Eligibility, Deadline)
- Application Table (ApplicationID, StudentID, JobID, Status)
- Interview Table (InterviewID, ApplicationID, Date, Venue, Result.

Chapter 8
Result

Results and Discussion

The Campus Placement Portal has been successfully implemented with complete development of all planned modules. The system includes student registration, profile management, job application functionality, admin dashboard, Excel database integration, authentication system, eligibility checking features, and notification capabilities. The Flask backend successfully handles routing, session management, and Excel file operations for storing student records, company details, job postings, and application data. The HTML, CSS ,and JavaScript frontend provides a responsive interface with separate student and admin views.Role-based access control works correctly, restricting students to their dashboard while giving admin full system access. Unit testing confirms proper functioning of individual components including login validation, data filtering by CGPA and branch, application submission to Excel sheets, and basic email notification generation. The complete workflow from registration to application submission has been developed and is ready for deployment. The system a waits institutional testing with live users for final validation and performance assessment in real-world campus recruitment scenarios.

1) Student login

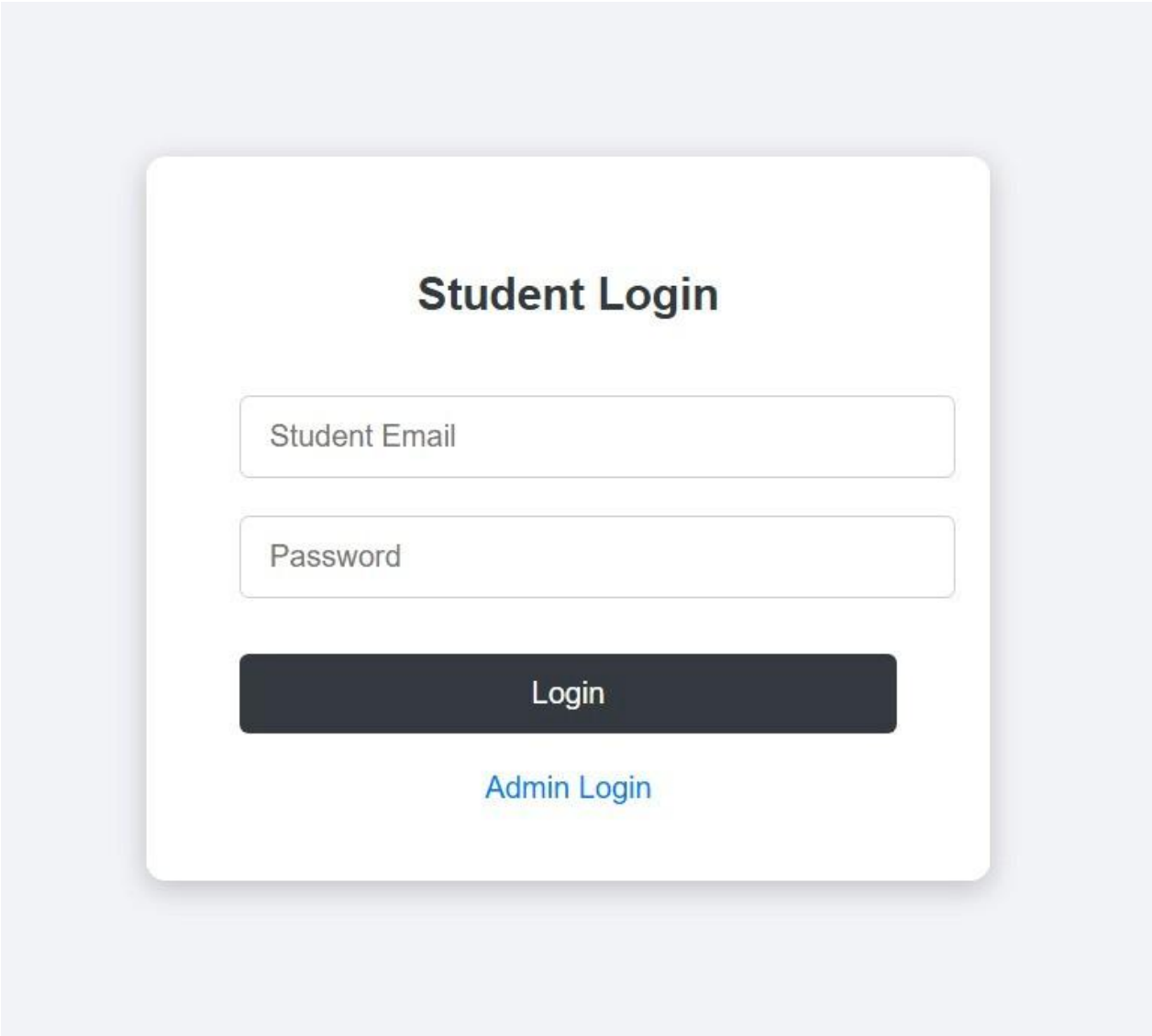

Fig 5.Student Login

This page is used by students to log in to the campus placement portal using their email ID and password. After successful login, the student can access their dashboard and placement features.

2)Student dashboard

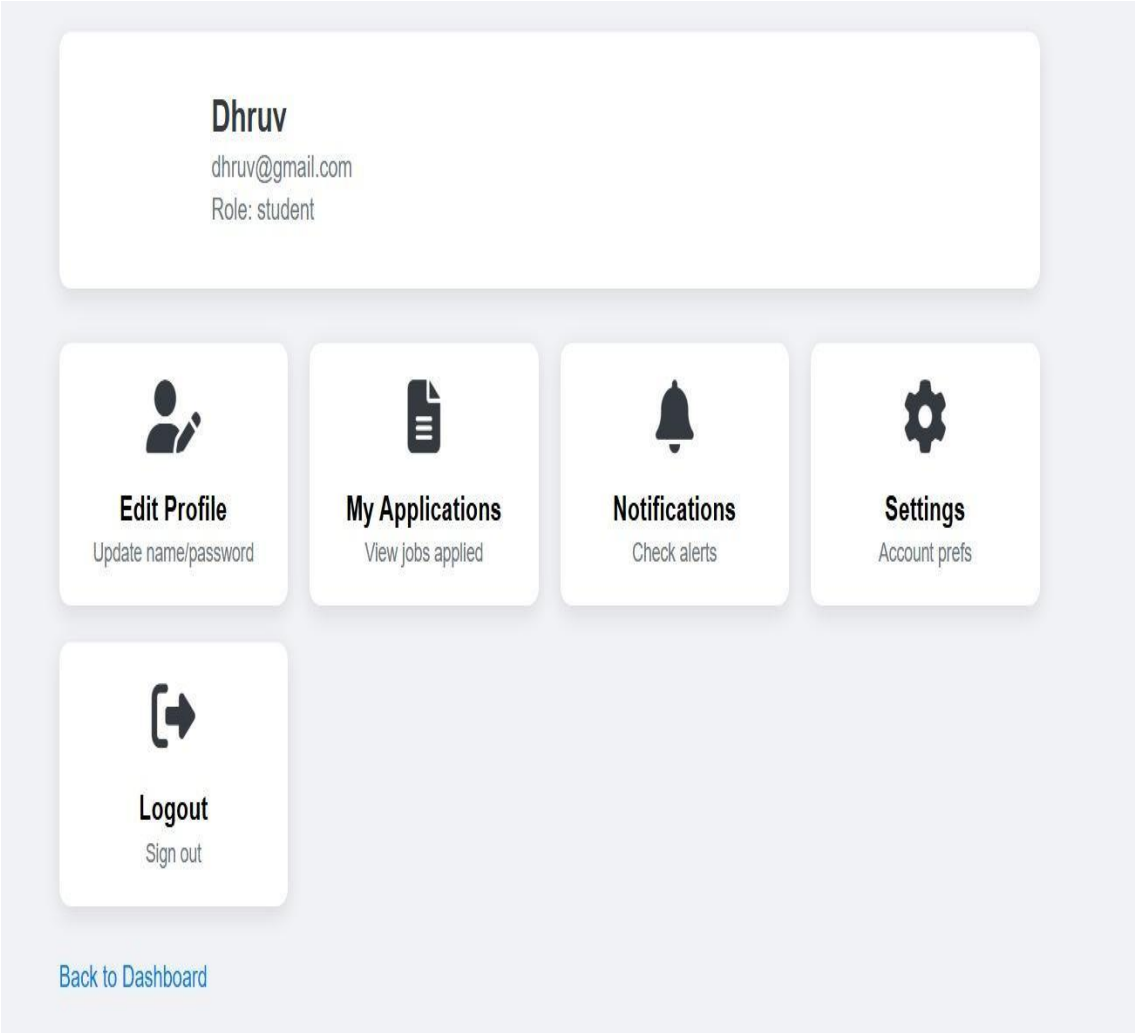

Fig 6. Student Dashboard

This is the main student panel after login. Here students can edit profile, view job applications, check notifications, change settings, and logout. It helps students track their placement activities easily.

3)Admin Dashboard

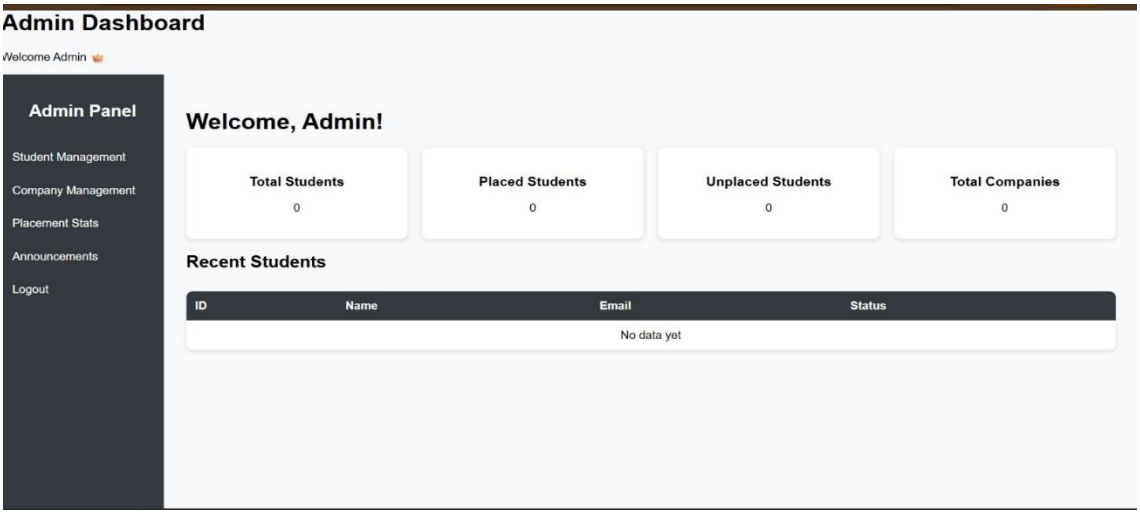

Fig 7.Admin Dashboard

This page shows the overall admin panel summary like total students, placed students, unplaced students, and total companies. Admin can also view recent student records and manage portal activities.

4)Add Company

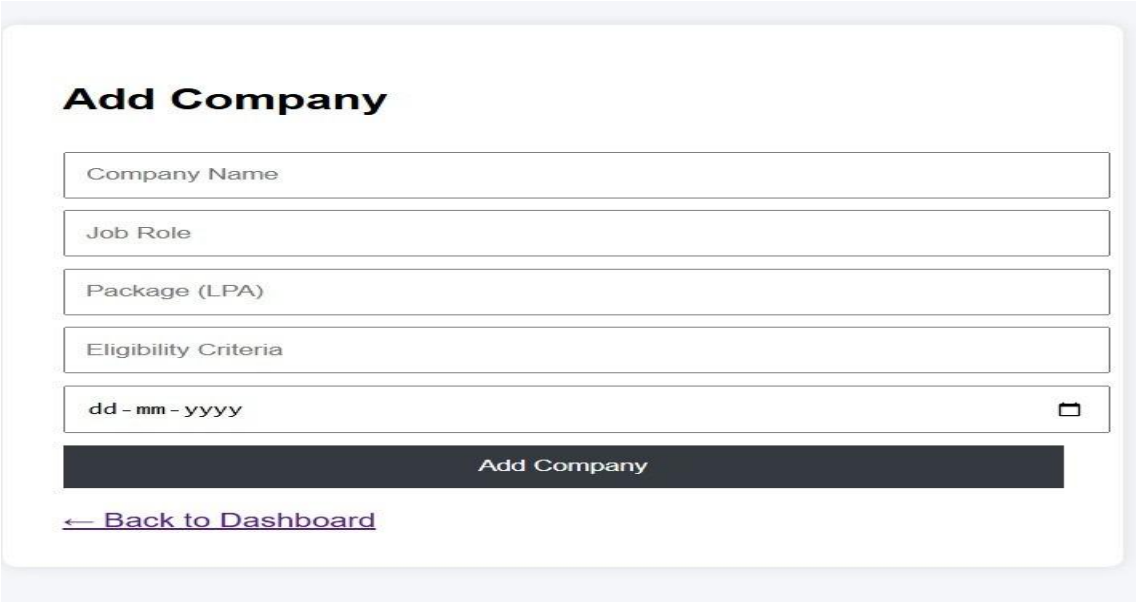

Fig 8.Add Company

This page is used by the admin to add new company details like company name, job role, package, eligibility criteria, and other requirements for campus placements.

5)Applied Companies

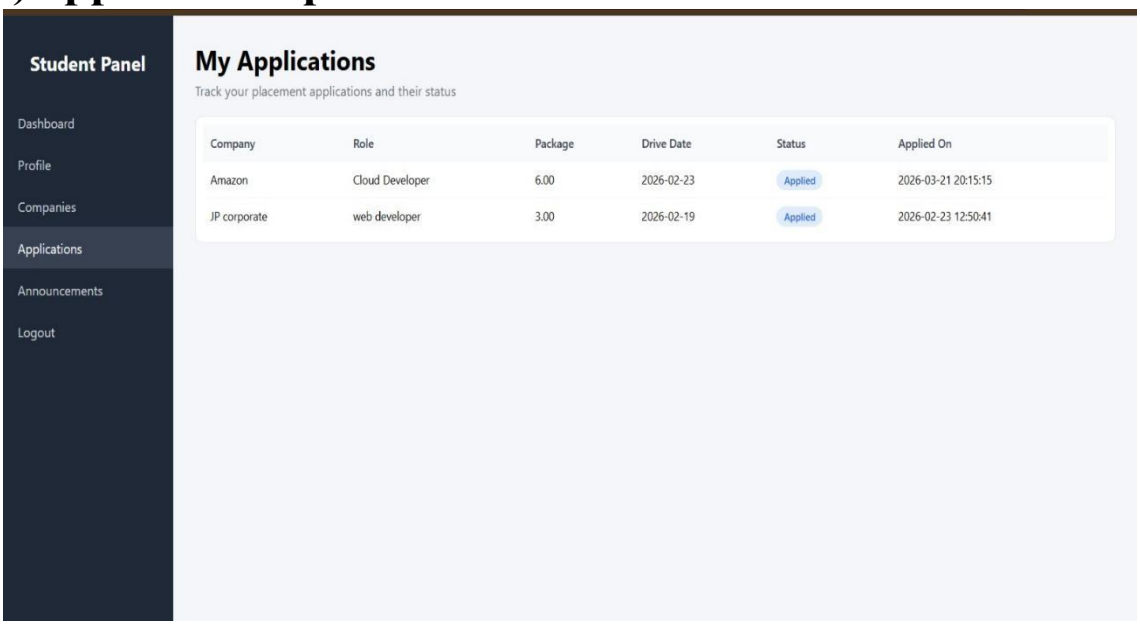

Fig 9. Applied Company

This page shows the list of companies applied by the student. Students can track their application status and view applied job details.

6)Placement stats

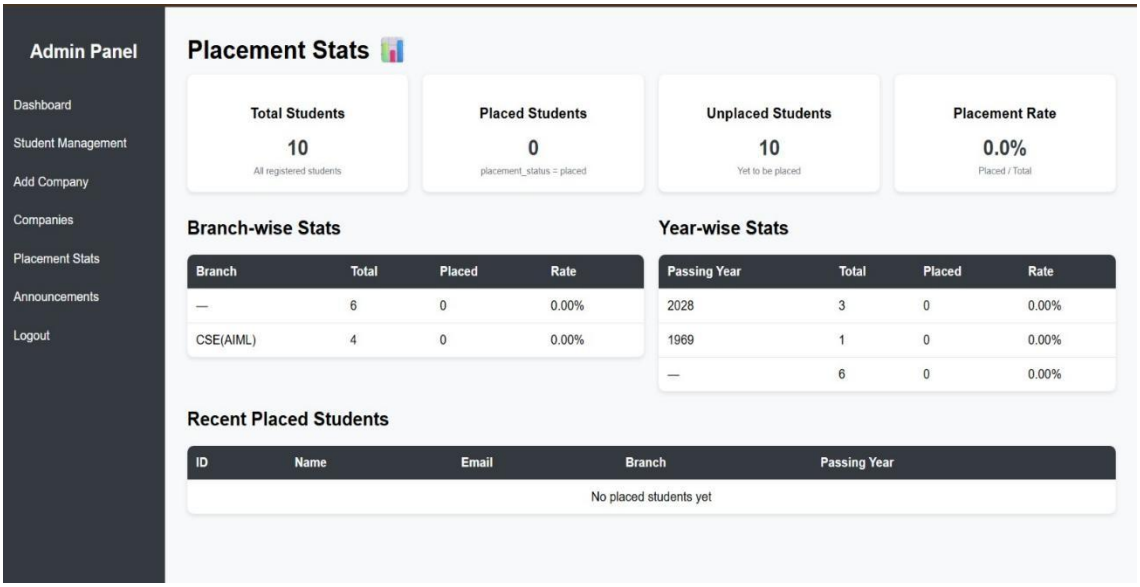

Fig 10. Placement stats

This page displays placement statistics like total students, placed/unplaced students, and branch-wise placement details. It helps admin analyze placement performance easily.

7)Excel Sheet

	A	B	C	D	E	F	G	H
1	Student ID	Name	Email	Branch	Passing Ye	CGPA	Semester	Pointer
2	1	Dhruv	dhruv@gm	CSE(AIML)	2028	7.9	4	7
3	12	Arya Hegis	arya@gma	CSE(AIML)	2028	9.04	4	9.04
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Fig 11. Excel Sheet

This sheet contains student data in tabular form such as Student ID, name, email, branch, passing year, CGPA, semester, and pointer. It is used for maintaining and managing student records easily.

Chapter 9
Conclusion

The Campus Placement Portal successfully addresses the limitations of traditional manual placement systems by providing a centralized web-based solution. Developed using Python Flask, Excel database, and responsive HTML/CSS/JavaScript interface, it automates student registration, job applications, eligibility checking, and admin management. The system eliminates paperwork, improves data organization, and ensures role-based access control for students and administrators. All planned modules have been implemented with verified unit testing, confirming functional Flask routing, Excel operations, and complete workflow development. The portal is ready for institutional deployment and live testing, demonstrating potential to streamline campus recruitment processes effectively while maintaining simplicity and reliability for educational institutions.

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